

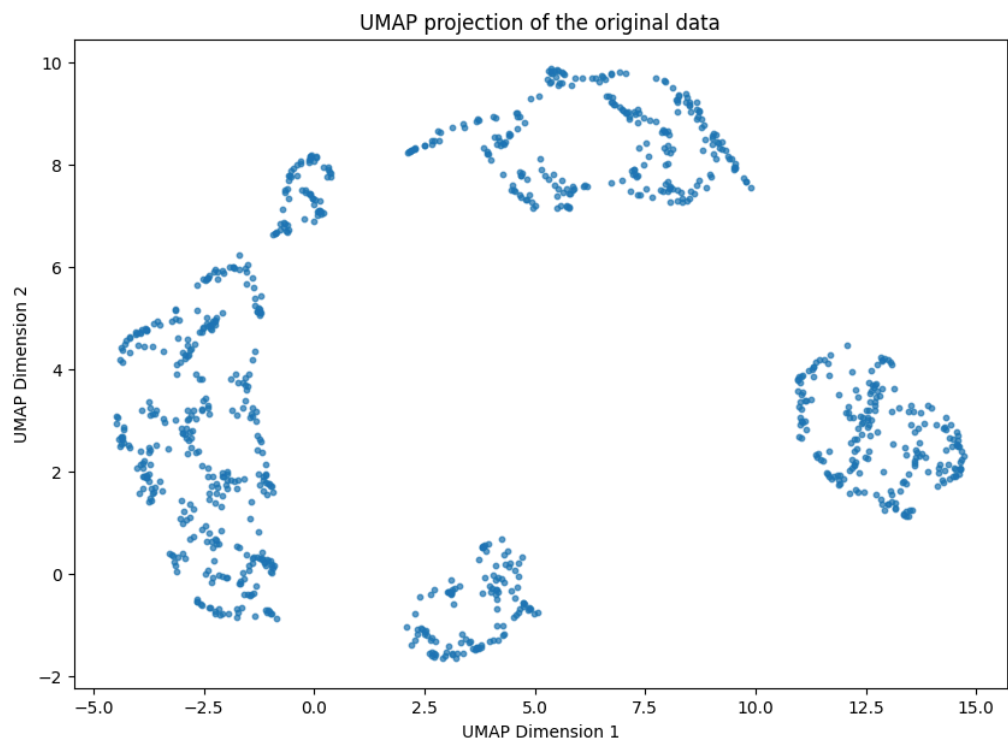
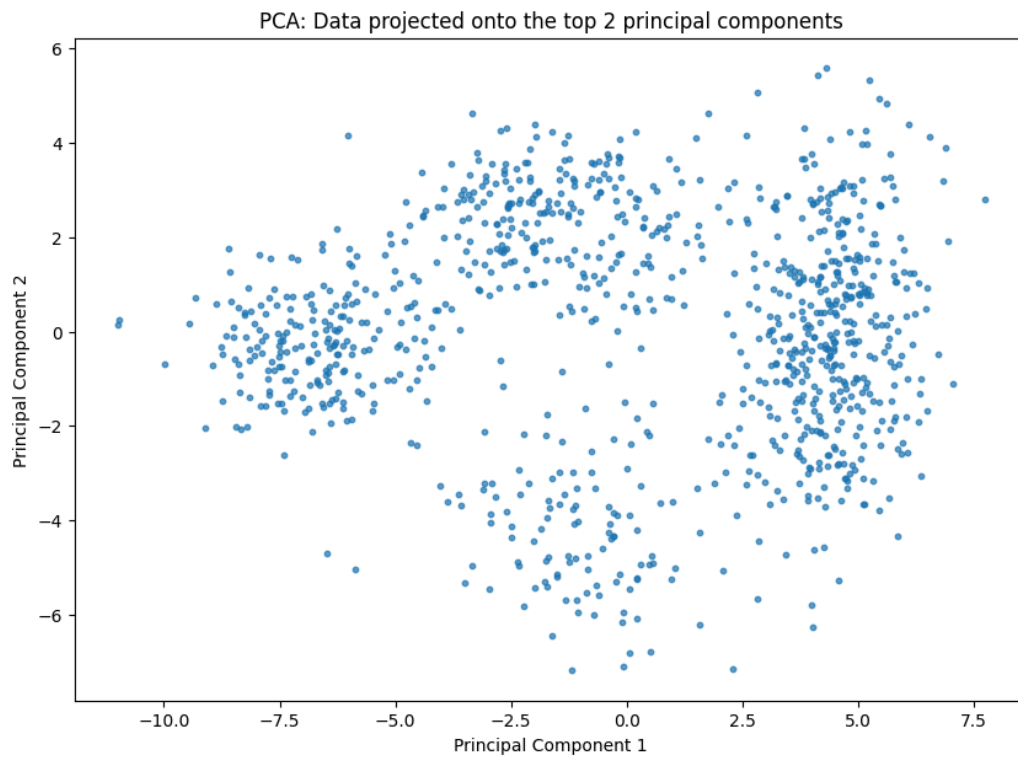


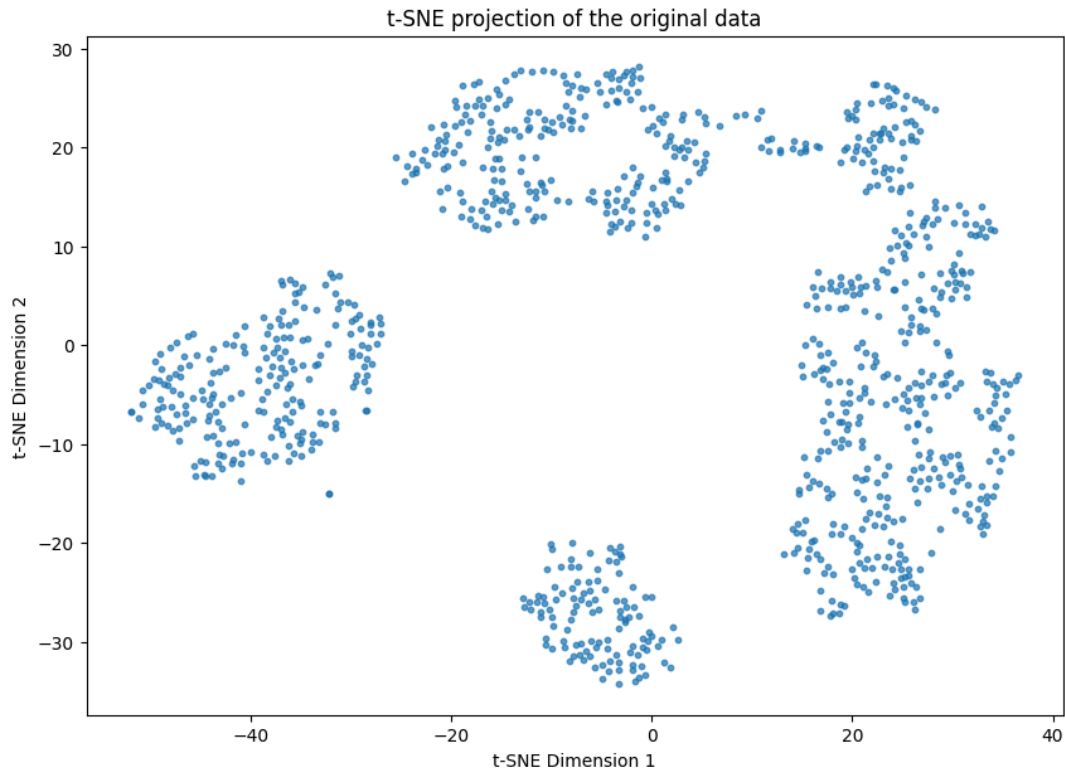
Assignment 4: Principal Component Analysis (PCA) And Expectation-maximization (EM) algorithm

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Principal Component Analysis(PCA)





EM Algorithm

Estimated mean number of children for families with family planning: 1.7839622667826571
Estimated mean number of children for families without family planning: 4.912037085130496
Estimated proportion of families with family planning: 0.35646113021025116
Estimated proportion of families without family planning: 0.6435388697897488

How To Run

It's a notebook file. If all the libraries are downloaded in the device, a simple run all in Python 3.12.0 kernel would be sufficient to run this code.